

EEGpal: **GEDAI** module

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Video tutorial: *Coming soon*

The GEDAI toolbox is a new way to correct EEG artefacts (muscles, eyes, electronic noise, etc..). **écrire un phase de résumé. based on the comparison of your data with a matrix of .**

This matrix corresponds to the theoretical EEG signal if all cortical sources were active at the same time. It is compute only based on the spatial distribution of the electrodes (synthetic signal) and not from real data.

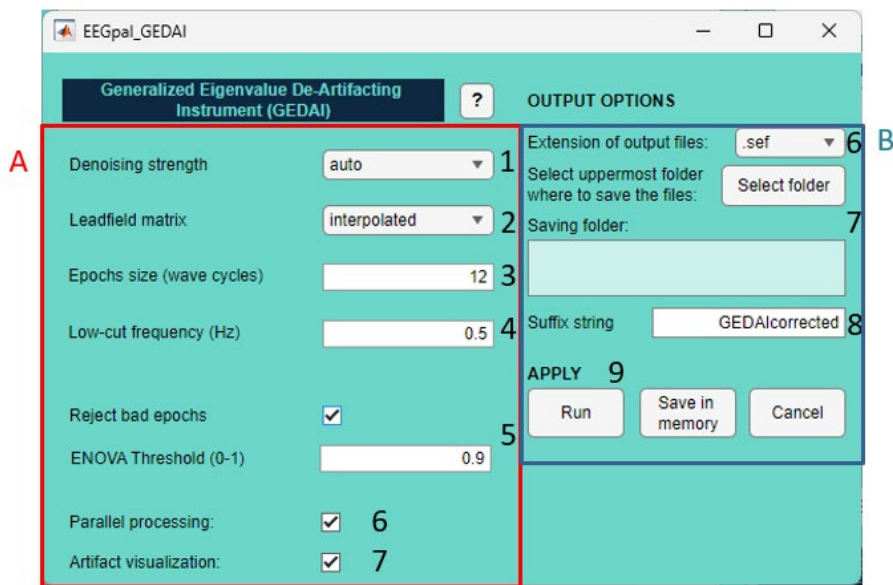
It replaces ICA or ASR and **should not be applied in addition to them.**

Multi-channel EEG may be considered to be a linear summation of electrical activities from a brain “signal” subspace with one containing different types of non-cerebral noise or “artifact”. This mixture may be “unmixed” by linear decomposition techniques (e.g. PCA or ICA) into separate components with individual source locations (“topographies”) and respective time-courses (“waveforms”). However, as “blind” source separation methods, PCA and ICA leverage statistical properties within mixed data to recover underlying sources, functioning without *a priori* knowledge of the original signals or their mixing process. GEDAI combines prior knowledge of the brain’s “signal” subspace (i.e. its spatial covariance) with generalized eigenvalue decomposition (GEVD) to more effectively separate source components belonging to the artifact-subspace from those of the brain-subspace.

A complete theoretical description and demonstration of its capacity can be found at the following website: <https://cibm.ch/noise-cancelling-headphones-for-the-brain-how-a-new-algorithm-eliminates-noise-in-eeeg-data/>

The original paper which described the algorithm can be found at the following location:

Ros, T., Férat, V., Huang, Y., et al. (2025). Return of the GEDAI: Unsupervised EEG Denoising based on Leadfield Filtering. bioRxiv. <https://doi.org/10.1101/2025.10.04.680449>



Pannel A:

1. Determining deartifacting strength.
Stronger threshold type:
 - a. **auto** : Default option:
 - b. **auto+** might remove more noise at the expense of signal, while milder threshold
 - c. **auto-** might retain more signal at the expense of noise.
2. The leadfield matrix corresponds to the matrix of the theoretical clean EEG signal used as a reference. This matrix has been pre-computed based on a configuration of 343 electrodes (10-5 layout). The toolbox needs to interpolate it to match the electrode configuration of your data using a spherical algorithm. By default, leave the option **interpolated**. Use the option **precomputed** only if the electrode names load in *Electrode Setting* (main EEGpal window) respect the international convention (e.g. Fp1, Cz, ...) with a standard electrode configuration.
3. Epoch size in number of wave cycles for each wavelet band. Default is 12.
4. Low-cut frequency in Hz. Wavelet bands below this frequency will be excluded. Default is 0.5 Hz to use only if no filter has been applied to your data. If you have already applied a high-pass filter to your signal using the Filtering+ module, this is unnecessary. In this case, it is recommended that you use the same cut-off frequency as in the Filtering+ module. For example, if you applied a high-pass filter at 1 Hz, you should set the low-cut frequency threshold in GEDAI to 1 Hz.
5. This option allows you to remove part of the signal rather than denoise it. Epochs are rejected based on Explained Noise Variance (ENOVA). Specifying a threshold of 0.9 will reject all epochs in which more than 90% of the signal is identified as noise. A smaller value will lead to more conservative processing. The toolbox will automatically adjust the position of EEG events recorded in MRK files.
6. This option allows you to speed up processing by using multiple threads. This option is only available if you have the **Parallel Computing Toolbox** from MathWorks installed in your Matlab.
7. Open an additional window overlaying the original and corrected signals for visualisation.

Pannel B:

8. Select the format for the output files.
9. Select the destination folder where the results files will be saved. Note that this reproduces the input structure. For example, if the input files were in subfolders, there would be a folder per participant.
10. The suffix added to the input filename to obtain the output filename.
11. There are three validation buttons:
 - a. The **Run** button will carry out the processing parameterized in the Re-referencing module.
 - b. The **Save in memory** button will store all the parameters in memory and close the Re-Referencing module without performing the processing.
 - c. The button **Cancel** closes the module without processing and without keeping the entered parameters in memory. The same effect will be achieved by closing the Re-Referencing module window.

FAQ

What is the impact of the parameter “Epochs size (wave cycles)”

In GEDAI, EEG data are not analyzed as one long continuous signal. Instead, the data are segmented into short epochs, and these epochs are used to compute frequency-specific covariance matrices that feed the GED (Generalized EigenDecomposition).

The parameter *Epochs size (wave cycles)* defines the length of each epoch in terms of the number of oscillatory cycles, not in seconds.

Key idea

The actual duration (in seconds) of an epoch depends on the frequency being analyzed:

$$\text{Epoch duration (s)} \approx \frac{\text{Number of cycles}}{\text{Frequency (Hz)}}$$

With the default value:

- Epochs size (wave cycles) = 12

Examples:

- At 10 Hz → epoch ≈ 1.2 s
- At 5 Hz → epoch ≈ 2.4 s
- At 30 Hz → epoch ≈ 0.4 s

GEDAI relies on:

- Band-limited covariance matrices
- Wavelet-based spectral filtering
- Generalized eigenvalue decomposition to separate neural signal from noise/artifacts

The epoch length directly affects:

1. Covariance estimation quality
 - Too short → ill-conditioned, noisy covariance matrices
 - Too long → smearing of nonstationary brain dynamics
2. Time–frequency trade-off
 - More cycles = better frequency precision, worse temporal precision
 - Fewer cycles = better temporal precision, worse frequency precision
3. Artifact detection sensitivity
 - Short epochs → better detection of brief EMG or movement artifacts
 - Long epochs → more robust detection of sustained noise sources

The default value (12 cycles) is a well-balanced compromise between statistical stability and temporal sensitivity across common EEG frequency bands.

Written by ChatGPT.